HighGen—Comparative genomics resources for cannabis

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Reference-guided whole-genome alignment pipelines are tuned for cross-species vertebrate panels. With **HighGen**, we adapted them in Galaxy across **217 *Cannabis sativa* assemblies** anchored on `cs10`—recalibrating repeat masking, empirical substitution scoring, chunked extension, and streaming 64-bit dynamic programming, paired with 43-genome Panagram *k*-mer bitmaps, a TOGA2 orthology pilot, and a 78-family phenotype ledger.

228-stanza (219-assembly) Hub
Attach depot.galaxyproject.org/highgen/hub.txt in the UCSC Genome Browser (Track Hubs → My Hubs).

A Four cross-species bottlenecks

In *Cannabis sativa*, interspersed repeats (~64–67% LTR retrotransposons) surround dense tandem metabolic arrays (Grassa et al. 2021, Lynch et al. 2025). Projecting 216 query assemblies onto the linear `cs10` (CBDRx) reference coordinate grid confronts four vertebrate-default obstacles:

1 Masking bias
Arrived soft-masking spans 0.00–80.86% across 218 assemblies, and KegAlign (Gulhan et al. 2025), in our measurement, neither seeds nor extends through lowercase bases

2 Cross-species matrix
HOXD70 (Chiaromonte et al. 2002) targets below 70% identity, which in *Cannabis* is 49.7% chance noise (Ts/Tv = 0.872 vs 0.74 floor)

3 Superlinear scaling
Whole-chr LASTZ (Harris 2007) scales as $O(n^{1.9})$, while 217-way MULTIZ (Blanchette et al. 2004) overflows 32-bit scores and buffers MAFs in RAM

4 Macro-syntenic nets
Human-mouse `netFilter -syn` (Kent et al. 2003, `maxFar = 200 kb`) prunes rearranged tandem gene clusters

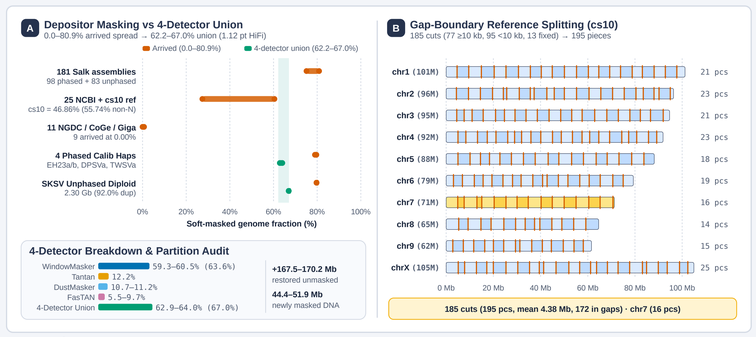


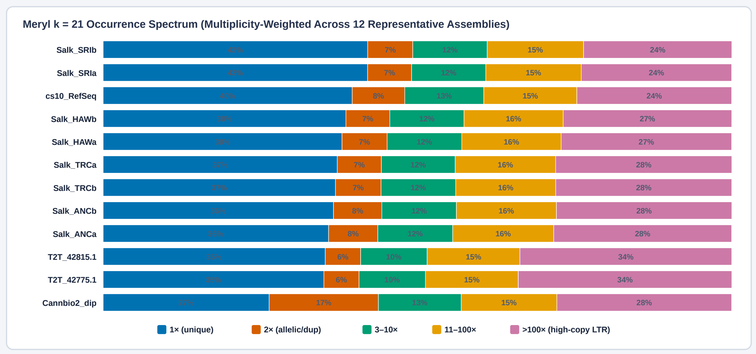
Figure 1 | Masking convergence and gap-boundary reference splitting. Stripping depositor masks across the 5-genome calibration panel and 5-genome intake pilot collapses a 0.0–80.9% intake spread into a 62.2–67.0% 4-detector union (**A**), while 185 chromosomal cuts (77 at N-runs ≥ 10 kb, 95 at N-runs under 10 kb, and 13 fixed 5 Mb cuts, `wfi_cs10_splits.bed`) partition `cs10` into 195 ~5 Mb pieces plus 26 heavy unplaced scaffolds, mitochondrion `NC_029855.1`, and `small.00`, with `mafSplit` cutting straddling blocks without base loss and `maf_concat` rejoining the pieces (**B**).

B Standardized 4-detector repeat masking

To eliminate submitter ascertainment bias across our 221 curated assembly records (217 non-redundant in MULTIZ— `cs10` plus 216 queries representing 164 prefix groups and 156 distinct plants, scored with `rosales_odb12` BUSCO, Lynch et al. 2025, Pike et al. 2025, 2026), our `brc-tools` UDT wrappers strip arrived masks and unite four de novo detectors (Morgulis et al. 2006, Frith 2011):

`fasta-uppercase` → `WindowMasker` `DustMasker` `Tantan` `FasTAN` → `bedtools merge`

Assembly / Cohort	Arrived Mask	4-Detector Union	Partition Audit and Compute
181 Salk assemblies (98 phased + 83 unphased)	75.12–80.86%	**62.19–67.04%** (5-genome intake pilot: 22.61 → 4.85 pt spread)	Depositors over-masked ~22% diverged unmasked DNA
25 NCBI GenBank + `cs10`	27.42–60.43% (`cs10` 46.86% total, 55.74% non-N)		Leaves 44.26% of `cs10` non-N bases open to seeding
11 NGDC / CoGe / GigaDB	0.00–0.68% (9 kept at 0.00%)		Prevents unmasked queries from seeding 100% of repeats
4 Phased Calibration Haps (`EH23a/b`, `DPSVa`, `TWSVa`)	**79.00–79.61%**	**62.90–64.02% (1.12-pt window)**	**WM 59.30–60.53%, Dust 10.7–11.2%, Tantan 12.2%, FasTAN 5.5–9.7% · 3.64–3.99 CPU-h/hap**
Diploid `SKSV` (2.30 Gb, 92.0% BUSCO dup)	79.71%	**66.95%** (4.05-pt panel window, WM 63.60%)	432.1 Mb restored, 139.1 Mb newly masked · 10.34 CPU-h



12-genome Meryl *k* = 21 occurrence spectrum: Phased haplotypes (`cs10`, 8 Pike et al. 2026 ONT, 2 IMPD T2T) carry 35–42% `1×` and 6–8% `2×` multiplicity-weighted *k*-mers, whereas uncollapsed diploid `Cannbio-2` (Braich et al. 2020) carries 27% `1×` and 17% `2×`.

Across phased haplotypes, the union restores 167.5–170.2 Mb of depositor-masked DNA while newly masking 44.4–51.9 Mb—a net unmasked gain of +115.9 to 123.2 Mb per haplotype with exact lowercase FASTA ↔ 0-based BED parity.

C Empirical intra-species substitution matrix

We adapted the Chiaromonte et al. (2002) log-odds protocol across two independent phased haplotype pairs (`EH23a × EH23b` and `DPSVa × TWSVa`, 95.3% `rosales_odb12` BUSCO completeness, 1.8–2.0% duplication). Inverting the training window to `[85, 95)%` identity, excising Salk GFF3 protein-coding CDS (–7.3% pre-chain segments), and enforcing single-chain colinearity (`--max-chains 1`) retained 54.5% of orthologous vs 6.6% median off-chromosome paralogous bases (**8.2-fold**) on `Cb2 × cs10` and **18.0-fold** genome-wide on `EH23a × EH23b`.

82.76 Mb
Physical `[85, 95)%` pairs (331.05M 4×-symmetrized, `Ts/Tv = 2.119`, H = 1.4140 bits, either pair alone within 1 unit)

10/10 Arms
Wins vs HOXD70 (`chr8` re-run after OOM, 9/10 loci p = 0.0107 vs α = 0.0083) and 67/70 windows (95.7% [91.4, 100%])

5,033×
`axtChain` shuffled-null suppression (`EH23a chr1 × Morus`) + scaled `682/51` gaps (70.6% single coverage)

Normalized to max match 100 (`+80` A/T, `+100` C/G, `−92` transitions, `−191` A-C/G-T, `−197` C-G, `−198` A-T transversions), our matrix also won **10/10 arms**, **10/10 loci**, and **62/70 windows** (88.6% [83.5, 94.6%]) vs unit `±100` and **55/70 windows** (78.6% [70.3, 85.5%], 9/10 loci) vs unchained across 40 false-match budgets—a consistent but small gain, with arm margins of 0.02–0.11% and no locus won without crossing curves.

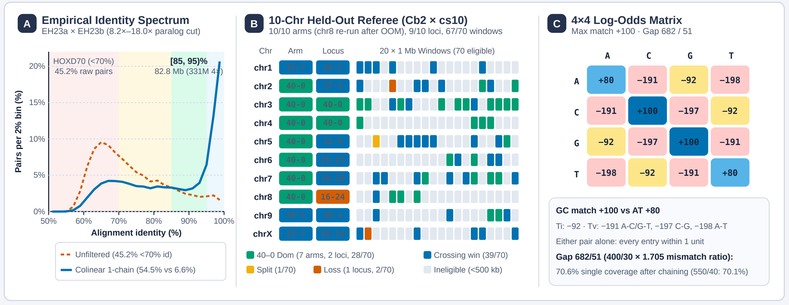


Figure 2 | Intra-species matrix calibration and 10-chromosome × 20-window held-out referee. Empirical identity spectra and single-chain colinearity separation (**A**), antisymmetric corner-threshold wins across 10 held-out `Cb2 × cs10` chromosomes and 70 eligible 1 Mb windows (**B**), and symmetric 4×4 log-odds weights with `682/51` gap scaling (**C**).

D Collection-native LASTZ and streaming MULTIZ 11.4

- **Growler LASTZ (`growler_ucsc_pair`, LASTZ 1.04.52/1.04.60):** Couples KegAlign v0.3.4 GPU collection seeding (Gulhan et al. 2025) with indexed `.2bit` archives, compact LAV, and piped PSL chaining (Harris 2007, Kent et al. 2003). Batching anchor files into `≤50,000`-segment chunks cuts `EH23a × EH23b` gapped extension **7.8-fold** (`389.0 → 49.8 slot-h`, `1.88 h` wall-clock, median 28 min/job, under `3.97` GB RAM) while matching **99.99%** of UCSC's `hg38 chr21 × panTro6 chr21 all.chain` and `rbest.chain` pairs and producing byte-identical `doRecipBest` chains across 5 test queries.
- **Storage and 218-run production cohort:** Purging transient intermediate collections after workflow completion reduced total Galaxy history storage across all 218 runs (54,554 jobs, 10,325 job-hours) from **14.96 TB to 6.50 TB**—reclaiming **56.5%** (`8.45 TB`). Syntenic nets cover a median **65.2%** of `cs10` non-N bases across 216 queries (range 12.2–85.1%, vs **70.0%** and 29.3–88.1% in `rbest`, with 40.0% diploid vs 54.0% phased median query-side coverage).
- **Streaming 64-bit column-count MULTIZ 11.3/11.4:** Replacing in-memory MAF buffering (v11.2) with coordinate-ordered streaming cursors and widening score accumulators to `int64_t` in v11.3 (`multiz#19`, fixing 29 deep blocks—24 with final scores $\geq 2^{31}$—on `NC_044376.1.02`, Blanchette et al. 2004) cut `hg38` 5-way `chrX` memory **51.7-fold** (`677 → 13 MiB`, 217-way worker RAM `≤14.1 MB`). Factoring `mz_yama.c` DP loops through column nucleotide counts (c_A, c_C, c_G, c_T, c_N) in v11.4 (`multiz#28`) accelerated a deep merge within a 216-genome roast **33.8-fold** (`981 → 29 s` against the 11.3 release build) and cut a whole 4.1 Mb piece **6.5-fold** (`7.3 → 1.1 h`, with 82 chr pieces merged under v11.3 and 113 + scaffolds under v11.4) with byte-identical output.

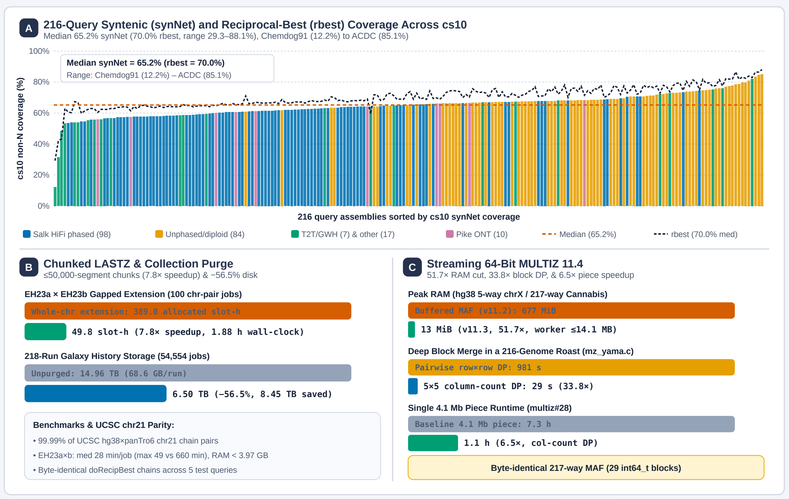
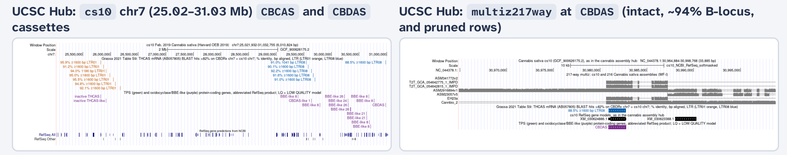


Figure 3 | 216-query syntenic coverage waterfall and scalability benchmarks. Sorted `synNet` (median 65.2%, range 12.2–85.1%) and `rbest` 1-to-1 (median 70.0%, range 29.3–88.1%) coverage across 216 queries by assembly class (**A**), chunked LASTZ speedup, `hg38 chr21` parity, and 218-run storage purge (**B**), and MULTIZ v11.3/v11.4 streaming RAM (51.7×) and column-count DP speedup (33.8× block, 6.5× piece) (**C**).

E Tandem metabolic clusters and syntenic net limits

- **Genome-wide tandem census and permutation audit:** Adapting MCScanX tandem criteria (Wang et al. 2012) to stricter BLASTp thresholds ($E \leq 10^{-10}$, top 10 hits), **3,035** of `cs10`'s 25,296 RefSeq protein-coding genes reside in 1,208 tandem arrays (K = 9) and **5,123 (20.3%)** in 1,796 clusters ($K \leq 10$), including **41 terpene synthases (TPS)** (31 in 9 clusters + 10 singletons) and **35 cannabinoid oxidocyclases (BBE-like)** (29 in 9 clusters + 6 singletons including `CBDAS`). Across 20,000 cluster-level permutations, nominal TPS low-identity enrichment (37.0% vs 11.0% controls, p = 0.022) does not survive family-wise correction (p_{FWER} = 0.12, threshold-free p = 0.40, BBE-like 2.8%, p = 0.59).
- **Functional `CBDAS` vs B-locus pseudogenes:** At `cs10` functional `CBDAS` (`LOC115697762`, chr7:30.98 Mb, Grassa et al. 2021), our 217-way alignment separates **79 assemblies** (59 plants, including 56 Lynch full-length `CBDAS`) at `≥98.5%` identity and **6 assemblies** below `93.0%` (`≥50%` CDS) from **77 assemblies** (67 plants) at `93.0–98.5%`—where all 63 Lynch-called contigs (62 THC-type, 1 both, 14 uncalled, Lynch et al. 2025) pair `cs10 CBDAS` with the ~94% B-locus pseudogene near `THCAS`.
- **`netFilter -syn` pruning vs paralog reuse:** Human-mouse distance and score cutoffs (`maxFar = 200 kb`, `minTopScore = 300,000` accounting for 14.5% of losses) pruned **4.7% of TPS** (418/8,856: 4.2% clustered, 6.4% singleton) and **10.7% of BBE-like** (807/7,560: 7.5% clustered, 26.2% singleton vs 5.2%/13.1% controls)—dropping 54 `CBDAS` assemblies (under `50%` CDS) in `synNet`, including Finola's intact 99.4% transposed copy (`CM011610.1:21.84 Mb`, 26.8 Mb from its enclosing gap). Removing `netFilter -syn` leaves **46.4%** of recovered pairs (569/1,225) with a query interval overlapping another family gene's, motivating complementary `rbest` nets and TOGA2 and TOAST/BOAST orthology (Hou et al. 2006, 2009).



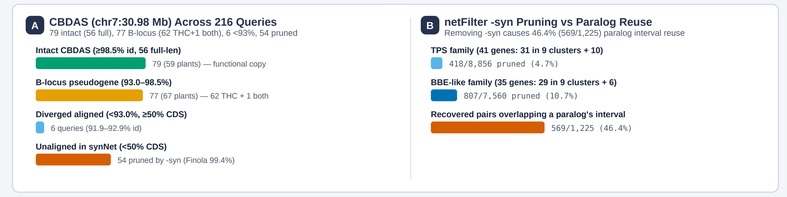


Figure 4 | Live HighGen UCSC Track Hub on `chr7` and `netFilter -syn` trade-off. Browser views of the 6 Mb cannabinoid synthase super-cluster (left) and `multiz217way` at `CBDAS` (right), paired with 216-query `CBDAS` resolution (**A**) and `netFilter -syn` pruning vs 46.4% paralog interval reuse (**B**).

F Panagram bitmaps, TOGA2 pilot, and phenotype ledger

- **Panagram v1.1.0 pan-*k*-mer indexing (`panagram_index` / `panagram_view`, Jenike et al. 2026, Kokot et al. 2017, Aylward et al. 2023):** Tier P1 (17 genomes: `cs10` + 16 Salk haplotypes from 9 plants, 13.3 Gb) selected k = 21 over k = 31 (higher linear 1 Mb window Pearson r with `rbest` identity in **16/16 Salk haplotypes**, mean r = 0.7150 vs 0.6901, Δr = +0.0249 within the 0.05 equivalence band). Tier P2 across **43 genomes** (32.64 Gb, 5.05 h on 16 Laila slots, 1.14 GB index) passed $r \geq 0.50$ in **41 of 42 chain-scored query genomes** (r = 0.5848–0.8118, including `Pink pepper` at r = 0.7376), passed non-PAR `chrY` separation (3/3 males 0.976–0.987 vs 3/3 V3 females ≤ 0.0773 across 205,619 100-bp-subsampled `AH3Ma`-absent positions), flagged 2 higher-QV Guppy 5 SUP trio-binned ONT haplotypes on `G3` count tolerance (`Punto Rojo` –10.4%, `Cherry Pie` –10.8%, Pike et al. 2025), and saw only one Sri Lankan ONT haplotype (`ASM5191173v1`, r = 0.4763, Merqury QV 24.01/24.11, Pike et al. 2026) fall below 0.50, for a cause still open.
- **TOGA2 v2.0.9 orthology pilot (Kirilenko et al. 2023, Malovichko et al. 2026):** On the `cs10` reference annotation, we stratified all **25,261** nuclear protein-coding genes (33,639 transcripts, excluding mitochondrion `NC_029855.1`) into **24,503 standard** and **758 reference-anomaly** genes alongside 221 inventoried `all.chain` datasets (1.45 TB). On its zero-divergence identity-chain control (`C-floor`), TOGA2 called **25 of 24,503 lost/uncertain (0.10%)**: 1 L + 24 UL against a ≤ 0.5% bar). On `EH23a`, the production, loose, and `RepeatFiller`-filled-and-cleaned chains called **2,077 (8.48%)**, **1,778 (7.26%)**, and **1,660 (6.77%)** standard genes L/UL. The filled and production arms differ on 713 of the 2,161 genes either calls L, UL, or M (33.0%), and production shows the least false presence (4,306 standard genes with more than one query gene, against 5,058 loose and 5,245 filled), so the filled arm does not beat it. The native-model check (`B6`) that ranks first is not yet implemented, so the pilot has chosen no arm and reports no gene loss.
- **Pedigree-verified phenotype ledger:** Joining 40 contributing sources—30 publications (29 paper texts, 2 supplements, and 988 Lynch et al. 2025 Figshare rows across 190 assemblies, from 43 audited paper extractions in `passed.tsv`) plus 10 NCBI BioSamples—across 215 assemblies yielded **1,341 trait records** across **154 plants in 78 pedigree families** (69 plants in the largest Salk family, 71 singletons). Under strict `measured` / `reported` filtering, 0 of 14 traits meet our ≥30-family, ≥2-class (≥5 families/class) association screen (`Varin` spans 30 families with `HO40` in 3 classes and 37/38 non-WT plants in `HO40`, whereas `Chemotype` reaches 30 families with 2 classes ≥5 if Lynch's HPLC method sentence covers its column).

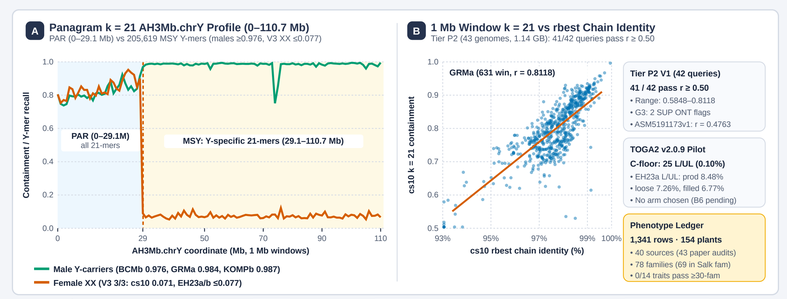


Figure 5 | Alignment-free Panagram `chrY` profile and `rbest` correlation. Chromosome-wide `AH3Mb.chrY` (0–110.7 Mb) k = 21 profile showing PAR all-21-mer containment (0–29.08 Mb) and MSY male-specific 21-mer recall across 205,619 subsampled positions (**A**), and all 631 eligible 1 Mb windows on `GRMa` (r = 0.8118) with Tier P2, TOGA2, and phenotype scorecards (**B**).

Open tools, workflows, and 228-stanza (219-assembly) browser hub

4-detector masking → `brc-tools` **GPU seeding + chunked LASTZ →** `growler_lastz_lav` **217-way streaming TBA →** `MULTIZ 11.4` **Pan-*k*-mer + orthology →** `Panagram` **and** `TOGA2`

Availability: HighGen 228-stanza UCSC Track Hub (219 unique assemblies across 157 plants + 9 alias stanzas: `multiz217way.bigMaf.bb`, reading frames, pairwise chains/nets, `rbest` chains, gene models, and repeat masks): `https://depot.galaxyproject.org/highgen/hub.txt` (each assembly's hub page credits its source, with 9 CoGe/NGDC redistribution licences pending per `assembly_licences.md`). Analysis repository, scoring matrix, and scripts: `https://github.com/richard-burhans/cannabis-genome`. Galaxy tool wrappers (`panagram_index`, `panagram_view`, `growler_lastz`, `gt-multiz-roast`) and workflows (`growler_ucsc_pair`, `wfi_split`, `wfi_roast`, `wfi_track`): `https://github.com/richard-burhans/galaxytools` (MIT) and `https://github.com/nekrut/brc-tools` · MULTIZ 11.3/11.4: `https://github.com/multiz/multiz` · LASTZ 1.04.52/1.04.60: `https://github.com/lastz/lastz` · KegAlign 0.3.4: `https://github.com/galaxyproject/KegAlign` · Panagram 1.1.0: `https://github.com/kjenike/panagram` · TOGA2 2.0.9: `https://github.com/hillerlab/TOGA2`.

HighGen Hub · Galaxy Tools · Repository